Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Concentration cells, types of concentration cells

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Concentration cells, types of concentration cells

1 Concentration cells, types of concentration cells

Study focus: Concentration cells, types of concentration cells. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000062 · EV0000063

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We'll continue with module 4 on electrochemical equilibrium.

Key details

- And in today's class we are going to talk about concentration cells, the different types of concentration cells, that is, electrolyte concentration cells and electrode concentration cells and then we study about ion selective electrodes.
- The different types of ion selective electrodes, and then we derive an expression for the electrode potential of an ion selective electrode.
- An electrochemical cell in which identical electrodes are in contact with the solution of identical species but of different concentrations.
- What does this mean?
- We know that the potential developed for this is going to be different from the one where the solution was of a dilute concentration.
- That we know from Nernst equation.
- electrodes and we have a salt bridge for electrical continuity, we find that the potential difference can be measured and this whole setup becomes of a galvanic cell.
- But each of these electrode is made of the same electrody material.
- The solutions are also of the same electrolyte, but the only difference is in the concentration of the.
- We have taken a cell which has two copper electrodes and they are immersed in copper sulphate solutions such that C2 is greater than C1.
- One of the concentrations is more than the other.
- And as usual, like any galvanic cell, oxidation will take place at the anode and reduction will take place at the cathode.

- The cell will be represented like this Cu, C U2 plus there's a soil bridge, and then we have again C U2 plus C U.
- The only difference is in the concentration.
- And similarly, at the cathode, we have C U2 plus taking up electrons and forming CO because there is a there has to be a reduction reaction again copper concentration is C2 at cathode is C2.
- Look at this reaction, we have to know what is Q.
- What type of electrode is this?
- I told in the previous classes that.
- This electrode is metal rod dipped in a solution of its own ions.

Example or application from the lecture

- Let us look at this picture.
- Let us take an example just now you saw in the picture.
- So let us say what are we assuming here that the concentration at the anode is C1 because remember cell representation, we always write the anode first and then the sol bridge and then the cathode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 05:00–09:59

For the anode would also be E naught plus 2.303 RTORNF log of C1.

Key details

- And E cell.
- When you write this expression we get E cell is equal to 2.303 R T over N F log of C2 by C1.
- What is N?
- The number of electrons getting exchanged and this is at.
- Room temperature that is 298 Kelvin.
- What we have to note here is that now E0 from both got cancel the anode and cathode e naught is not there.
- So therefore the expression does not contain e naught.
- The cell potential is going to be positive only when C2 is greater than C1, because.
- The expression was 0.0591 over n log of C2 by C1.
- We know that we took.
- C2 as the cathode concentration and C1 as the anode concentration.
- You can see here, here the E cell will be positive.
- Positive provided the cathode concentration is more than the anode concentration.
- What does it mean?
- If the difference in the.
- Concentrations is very high, then the power which is generated, the EMF, is going to be high.

- If the concentration is not very high, we do not get a very high value of EMF because of the.
- Again, because of the expression where the ratio C2 by C1 is appearing.
- Therefore, if C2 if the ratio is high, then only we get the a good power.
- We can we have a high EMF being generated from this set.
- That the cell potential will be 0.
- When C1 is different from C2, and when will it be positive if C2 is greater than C1.
- This third interpretation, the third point which we discussed just now, when C2 is equal to C1, EMF of the cell becomes equal to 0.
- Finally, the cell wants to come to a condition where EMF of the cell becomes E is zero because that also means.
- Delta G is equal to 0, and what does that mean?
- That the cell has reached equilibrium.
- So what would be the equilibrium condition for a concentration cell?
- C2 becomes equal to C1.
- And since we are saying C2 is a higher concentration, it's very obvious that higher concentration will try to lower the concentration, and that is possible only if a reduction reaction takes place.
- The other case where at the anode we said the C1 concentration is low concentration.
- How will it increase only if the metal dissolves more and comes into the solution.
- Therefore, at the cathode, the concentration should be.
- high as compared to the anode and once these concentrations are equalized then the cell PMF becomes zero.
- So in other words the cell has reach an equilibrium condition.
- As long as there is a difference in the concentration the cell is going to function.

Example or application from the lecture

- is always E cathode minus E anode and all you have to remember here is that we always consider the reduction potential.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 10:00–14:59

Just now what we saw was an electrolyte concentration cell.

Key details

- There's another type of concentration cell which is known as.
- The electrode concentration cell.
- We have the electrolyte concentration cell where we have two same electrodes that are dipped in the same electrolyte but with different concentrations of the electrolyte.
- Would again, since it is a concentration cell, the electrodes are same, the electrolyte is same, but here.
- The activity of the electrode itself is different.

- The concentration of the electrolyte remains the same, but the activity of the electrode itself is different.
- I taught you about sodium amalgam in types under types of electrode.
- This is another type of electrode where the metal is taken in the form of amalgam to reduce the activity of the metal.
- These two electrodes now what is different?
- The electrolyte concentration, if you see there's only one electrolyte.
- The activity of the electrode itself is different.
- at the anode divided by the concentration of the sodium amalgam at anode divided by the concentration of the sodium amalgam at the cathode.
- How do we get to that expression?
- Two sodium amalgam electrodes which are coupled together and both are dipped in a solution of Na plus ions.
- What would be the reaction at anode and cathode?
- We have C2 activity as you can see from the cell representation.
- That's an oxidation reaction.
- At the cathode we have NA plus electron is giving you sodium amalgam.
- Of C2 concentration.
- We are writing the cathode.
- H G C2 divided by NA plus.
- And similarly, the Nernst equation for the anode for the reduction potential will be E naught minus 2.303 R T over N F log of sodium amalgam with C1 concentration by NA plus.
- E right minus e left, that is e cathode minus e anode, and finally you get this expression.
- Then now you see 2.303 R P R N F log of NAHG C1.
- It's the anode concentration and C2 is the cathode concentration.
- It is not that always we have the cathode concentration will should be higher than the anode concentration.
- at the one at the cat hole.
- That for every system we need to derive and see.

Example or application from the lecture

- Just now we saw this example where we have two copper electrodes but of different concentrations of copper sulphate and when they are coupled together we see that EMF is generated and the cell potential is given by 2.303 R T over N F log of C2 by C1.
- Let us concentrate on that.
- Where what is an example for such an electrode?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 15:00–19:59

What is the final expression?

Key details

- Only then can we suggest whether the anode concentration should be higher or cathode concentration should be higher.
- It is what do we understand?
- It is based on the reactions happening at the anode and cathode.
- Such an electrode concentration cell would be if you have two hydrogen electrodes, you can see here.
- Electrolyte concentration cell, it is an electrode concentration cell because the electrode here, the hydrogen itself has a different partial pressure as compared to the hydrogen at the.
- They are at different partial pressures.
- Electrode concentration cell.
- Again, here we see it is the and the cell potential will be positive provided the.
- What is P1?
- The partial pressure of hydrogen at the anode is higher than that of the cathode.
- That the anode concentration should be high or the cathod concentration should be high.
- If we take a chlorine electrode, like two chlorine electrodes, now obviously the electrolyte is common, so there it is again an electrode concentration cell.
- You are in a log of P2 by P1.
- It is different from what happened for the hydrogen electrode.
- We have to derive and find out what the expression.
- Cathode partial pressure should be higher of chlorine as compared to the partial pressure at the anode.
- We need to determine what is the E cell, what is the expression for E cell, in order to understand when the cell reaction is going to be spontaneous or when E cell will be.
- positive and based on that we can decide the cathode and the anode concentration provided we want a positive emf then we have to see the anode and the con cathode concentration we will decide based on the expression which we get.
- Very selective to a specific ion, and that is why they are so important.
- What does it mean?
- That in the presence of other ions, it can selectively detect one particular ion, not only selectively detect, it can actually tell us.
- The concentration that means it can qualitatively detect the ion ignoring the other ions and it will also tell quantitatively tell us the concentration of that particular ion.
- Ion selected electrodes that is how it is named, and therefore the potential will be since it's an electrode, there will be a potential form, and that is going to be a function of the concentration of that particular ion.
- It is determining concentration of one iron in the presence of other ions, so it can actually act as a chemical sensor.
- They have a membrane.
- How do they achieve that?
- That finding only one ion when there are so many other ions present, it is because of the membrane which is capable of exchanging only that specific ion.
- The most important part is role is played by the membrane.
- It is capable of exchanging the specific ion with the solution with which it is in contact.

- Being able to recognize and detect and also quantitatively tell us the concentration of that particular ion, it is because of the membrane.

Example or application from the lecture

- So another example for.
- So this is again another example for.
- The Nernst equation we can derive and find out that it will be given by 2.303 R T over N F log of P1 by P2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 20:00–24:59

Glass electrode is an example for a membrane electrode.

Key details

- They are based on the membrane which is being used.
- As I said, membrane is the most important part of the ion selective electrode.
- Being used, there can be different types of ion selective electrodes.
- So the first type would be where we have crystalline.
- We find that this.
- This is selective to fluoride ion.
- It is lanthanum fluoride.
- in the presence of other ions and where would this be used in water pollution there is a lot of fluoride contamination and that is very dangerous.
- Such an ion selective electrode will be used to determine the fluoride ion concentration in the presence of other ions.
- This was single crystal, this is polycrystalline, where you have AG2S which is being used as a membrane, and that is selective to sulfide ions.
- We know that glass electrode, what does it have?
- It has a glass membrane.
- And this glass membrane happens to be selective to H plus ions mostly, but it can also be selective to sodium ions also.
- So glass membrane is the most important constituent of the glass electrode, and since selective to H plus ions, that means.
- We can determine the pH of a solution using a glass electrode.
- And then we can also have immobilized liquid in a rigid polymer.
- that iron exchangers as the name suggests are capable of exchanging ions to this would be very useful in ion selective electrodes so we have electrodes which are capable of detecting calcium and nitrate ions.
- Why is it a function of the concentration of the ion?
- The electrode potential of an ion selective electrode.
- Ion selective electrode.
- where the concentration is C1.

- Concentration of the Mn plus ions which I want to determine using this ion selective electrode.
- Come to the other side, we have the we have a reference solution.
- Determine the concentration.
- It will be the same solution on this side, but this concentration is known to us.
- Since we are calling it a reference solution, internal reference solution, we understand that this concentration is known to us.

Example or application from the lecture

- The best example for this is single crystal electrodes where we have which are selective to fluoride ions.
- For example, glass.
- Let us understand the principle behind how are how is the electrode potential getting formed in an ion selective electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 25:00–29:59

Is what I want to determine.

Key details

- And apart from this, there is also an inner reference electrode.
- We know it is because of the difference in the concentration across the membrane.
- Therefore, the juntial expression will be 2.303 RT over NF.
- Log of C1 by C two.
- Log of C1 and C2 has gone into the C2 has now become the constant.
- K will be.
- Minus, sorry, minus is missed over here, minus 2.303 R T over NF log of C2.
- But if you look at the membrane itself, the total membrane potential is going to be the sum of the junction potential plus the e reference electrode, because we have placed an reference electrode also which is a part of the ion selective electrode.
- What is the role of this reference electrode, the internal reference electrode?
- Moreover, the junction potential.
- It is there, but there is no external contact, no electrical contact by which we can measure this potential.
- Our aim is to find out what is this junction potential.
- Because why we know that junction potential will tell me what is the concentration C1.
- The moment I say reference electrode, I know what is the potential of the internal reference electrode.
- E reference that is the internal reference electrode potential.
- So therefore the role of this reference potential is just to provide me an electrical contact so that I can.
- Find out what is this junction potential.

- substituted here and plus the e reference.
- Pertaining to that particular electrode which is being used.
- And what is E0 M?
- K plus E reference.
- E m is equal to E naught M plus 2.303 R E over N F log of C1, where E0 M is equal to K plus E reference.
- We still cannot determine C1.
- Because this is a single electrode, and that single electrode potentials cannot be measured.
- We are going to club this electrode with another reference electrode.

Example or application from the lecture

- The junction potential is given by 2.303 R T over NF.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 30:00–34:59

That will be known as an external reference electrode.

Key details

- Because we have already used a reference electrode which is part of the in ion selective electrode.
- To make a cell using the membrane electrode, I need another reference electrode, and that will be referred to as the external reference electrode.
- Here it is the membrane electrode and then here on the anode we have the external reference electrode.
- And this is constant or known to us.
- So therefore, from here I can determine what is E membrane.
- E membrane is the membrane potential, and once I know membrane potential, I can always find out what is.
- So therefore, when we are measuring using any potential meter, so basically it is all automated whenever you want to measure the.
- We can know what is the concentration, but what is happening inside is actually the cell potential is being measured and it is being subtracted, it is being.
- subtract the membrane electrode is calculated using this expression.
- And we can get the concentration of the species directly when the electrode is dipped into the solution.
- The membrane which is being used offers a very high resistance.
- Membrane offers a very high resistance and therefore we will we cannot use ordinary potential meters, therefore, special type of potential meters have to be used whenever.
- We want to measure the cell potential where one of the electrode is an ion selective electrode.
- So, because the membrane which we are using offers very high resistance.
- These ion selective electrode they are used to determine concentration of number of cations and anions, such as we have hydrogen ion, we have lithium ion, sodium ion, potassium, etc.,

lead and arsenic also, and then we can also determine anions.
- like cyanide, nitrate, fluorides or sulfide.
- So, in the next class, we are going to study a specific ion selective electrode that is the glass electrode, which is the most profusely used ion selective electrode.

Example or application from the lecture

- Concentration directly concentration is given to us.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Concentration cells, types of concentration cells transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Concentration cells, types of concentration cells transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00015	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00017	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice